

Supplementary Materials

Supplement Materials TABLE I

ROBUSTNESS EVALUATION ON CIPHERSPECTRUM UNDER MIXED ATTACK (STR = 1.0, %)

Methods	Rob_Acc	Rob_F1	ASR	Rel_Drop	ECE
USTC-CNN [32]	24.02	25.69	55.57	52.43	26.52
DFE [36]	7.82	7.13	90.59	88.53	62.95
ET-BERT [37]	66.09	71.95	12.33	11.63	3.82
FS-Net [35]	61.72	61.52	31.72	29.37	19.75
SMGF-Net	48.90	48.24	44.19	41.55	24.72

Supplement Materials TABLE II

ROBUSTNESS EVALUATION ON USTC-TFC2016 UNDER MIXED ATTACK (STR = 1.0, %)

Methods	Rob_Acc	Rob_F1	ASR	Rel_Drop	ECE
USTC-CNN [32]	60.87	51.48	38.64	37.35	26.31
DFE [36]	33.63	30.40	66.37	63.31	48.81
ET-BERT [40]	97.13	84.39	1.76	1.61	1.33
FS-Net [35]	41.82	55.26	58.21	54.27	41.44
SMGF-Net	55.71	56.08	41.09	40.39	31.51

Supplement Materials TABLE III

ROBUSTNESS EVALUATION ON CIPHERSPECTRUM UNDER ADAPTIVE ATTACK (STR = 1.0, %)

Methods	Rob_Acc	Rob_F1	ASR	Rel_Drop	ECE
USTC-CNN [32]	37.87	39.64	29.85	25.02	16.73
DFE [36]	7.62	6.88	91.09	88.81	62.98
ET-BERT [37]	60.73	67.87	19.77	18.79	3.75
FS-Net [35]	62.87	62.62	30.45	28.05	18.49
SMGF-Net	71.44	71.20	17.43	14.62	4.76

Supplement Materials TABLE IV

ROBUSTNESS EVALUATION ON USTC-TFC2016 UNDER ADAPTIVE ATTACK (STR = 1.0, %)

Methods	Rob_Acc	Rob_F1	ASR	Rel_Drop	ECE
USTC-CNN [32]	57.4	44.23	41.28	40.93	32.35
DFE [36]	33.42	29.64	66.21	63.54	49.11
ET-BERT [37]	91.81	77.63	7.22	7.00	4.85
FS-Net [35]	40.97	37.13	56.22	55.21	42.81
SMGF-Net	73.89	74.35	21.57	20.94	11.23

Supplement Materials Proof A: Mathematical Proof of Modality Poisoning Isolation

To rigorously demonstrate how the semantic-guided dynamic gating mechanism mitigates modality poisoning, we present a mathematical comparative analysis between static fusion and the proposed dynamic gated fusion under adversarial structural perturbations.

A.1 Problem Formulation and Threat Model

Let $h_G \in R^{d_G}$, $h_{stat} \in R^{d_s}$, and $h_{sem} \in R^{d_E}$ denote the extracted embeddings for the macroscopic graph topology, global statistics, and micro-semantics, respectively.

Under a semantically constrained evasion attack, the adversary manipulates the physical traffic behaviors (e.g., dummy packet injection), applying an adversarial perturbation δ_G to the graph modality. The corrupted topological representation is formulated as:

$$h'_G = h_G + \delta_G$$

Concurrently, the micro-semantic features remain strictly bounded by protocol constraints, yielding invariant semantic representations ($h'_{sem} \approx h_{sem}$).

A.2 Vulnerability Analysis of Static Fusion

In conventional multi-modal frameworks, static fusion concatenates representations and projects them via a static weight matrix W_{static} . We can decompose W_{static} into sub-matrices corresponding to each modality: $W_{static} = [W_G \parallel W_{stat} \parallel W_{sem}]$. The output Z_{static} is given by:

$$Z_{static} = W_G h_G + W_{stat} h_{stat} + W_{sem} h_{sem}$$

When the adversarial perturbation δ_G is introduced, the corrupted output Z'_{static} becomes:

$$Z'_{static} = W_G (h_G + \delta_G) + W_{stat} h_{stat} + W_{sem} h_{sem}$$

The propagation of the adversarial noise through the static fusion channel is calculated as the difference ΔZ_{static} :

$$\Delta Z_{static} = Z'_{static} - Z_{static} = W_G \delta_G$$

Taking the norm, we obtain the magnitude of the propagated perturbation:

$$|\Delta Z_{static}| \leq |W_G| |\delta_G|$$

Since W_G is a fixed, optimizable weight matrix learned during training, it remains constant during inference. Therefore, the structural noise δ_G propagates linearly and unconditionally into the global representation, leading to catastrophic modality poisoning.

A.3 Proof of Noise Isolation via Semantic-Guided Dynamic Gating

To suppress this linear propagation, SMGF-Net constructs an invariant global context c_{global} and generates a dynamic credibility gate $g(x_{sem}) \in [0,1]^{d_G}$ driven by the unalterable micro-semantics. The gated fusion output is formulated as:

$$Z_{gated} = W_f [(g(x_{sem}) \odot h_G) \parallel c_{global}]$$

where W_f is the projection matrix and $\odot$ denotes the element-wise Hadamard product. Under the identical adversarial perturbation δ_G , assuming the semantic anchors successfully identify the topological anomaly, the gate dynamically calibrates to a penalized state g . The perturbed output is:

$$Z'_{gated} = W_f [(g \odot (h_G + \delta_G)) \parallel c_{global}]$$

Since the semantic-derived context c_{global} remains invariant, the perturbation propagation ΔZ_{gated} is isolated to the gated topological channel:

$$\Delta Z_{gated} = Z'_{gated} - Z_{gated} = W_f [(g \odot \delta_G) \parallel 0]$$

Applying the sub-multiplicative property of matrix norms, we establish the strict upper bound for the noise propagation:

$$|\Delta Z_{gated}| \leq |W_f| |g| |\delta_G|$$

Unlike the static constraint $|W_G|$, the dynamic gate norm $|g|$ acts as an active non-linear filter. When the network detects severe structural pollution via invariant semantic anchors, the

credibility score drops, forcing $|g| \rightarrow 0$. Consequently, the upper bound of the perturbation propagation is strictly restricted:

$$\lim_{|g| \rightarrow 0} |\Delta Z_{gated}| = 0$$

This mathematical formulation proves that the semantic-guided dynamic gating mechanism actively isolates adversarial noise, guaranteeing the purity of the global decision boundary even under severe topological corruption.

Supplement Materials Proof B: Mathematical Proof of Decision Boundary Robustness

Consider the latent representation z_i of class y . Define the intra-class radius as

$$r_y = \max_i |z_i - \mu_y|. \text{ Let } m_{y,y'} = |\mu_y - \mu_{y'}|$$

denote the margin between class y and y' . Under adversarial perturbation, the latent representation shifts by Δz .

Robust classification requires the perturbed representation to remain within the decision region of class y :

$$|(z_i + \Delta z) - \mu_y| < \frac{1}{2} m_{y,y'}$$

Applying the triangle inequality:

$$|(z_i + \Delta z) - \mu_y| \leq |z_i - \mu_y| + |\Delta z| \leq r_y + |\Delta z|$$

The condition for robustness becomes:

$$r_y + |\Delta z| < \frac{1}{2} m_{y,y'}$$

Supervised contrastive learning directly minimizes the distance between z_i and the class centroid μ_y . This reduces r_y . It simultaneously pushes different class clusters apart to increase $m_{y,y'}$. The ratio of margin to radius increases. The model resists adversarial crossing of the decision boundary.